

# How Many Independent Models Does an Open-Model Ecosystem Contain? Evidence at the Exact Conditional Null

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## Abstract

Whether independently developed language models fail on the same inputs bears directly on systemic risk, ensemble reliability, and model-as-judge evaluation. Recent work has shown that the answer depends on the null model chosen, and concluded that monoculture inference is therefore *subjective*, leaving open which null is appropriate [Jo et al., 2026]. We argue that one null is canonical rather than arbitrary. Because the row and column margins of a model-by-item outcome matrix are the jointly sufficient statistics of the Rasch family, the uniform distribution over margin-matched matrices is the unique *exact, fit-free* conditional null for that entire family: it estimates no parameters, applies no penalty, and makes no identification choice. We report what the answer is at that null, on 1,228–1,373 open models per benchmark across four benchmarks — a population three times larger than previously analysed.

At the mean level there is nothing to find, and this is a theorem rather than a measurement: mean pairwise co-failure is a function of the item margins alone, so its excess over any margin-preserving null is identically zero. We show this degeneracy has been proved independently in three literatures — as Schluter’s V-ratio in ecology, as Kuder–Richardson / Cronbach’s  $\alpha$  in psychometrics, and as Fleiss’s observed agreement for the multi-category case — and give the translation between them, since none is cited in the model-evaluation literature. Empirically the measured excess matches the closed form to  $10^{-16}$  on all four benchmarks. Our pre-registered dispersion statistic came out *below* its null on three of four benchmarks and above on one; its pre-registered kill condition fired and we report it as refuted.

What survives is second-order and large. Conditioning on the margins leaves residual correlation with root-mean-square 0.248 against 0.039 under the exact null — a sixfold excess — and five eigenvalues above the null spectral edge. We deliberately do *not* report this as an “effective number of independent models”: we show that the participation ratio is algebraically  $N/(1 + (N - 1)\bar{R}_{ij}^2)$ , so it cannot distinguish one weak global factor from many tight clusters, and we give the diagnostics that can. Misspecification controls show that two-parameter discrimination heterogeneity cannot generate the observed effect (rms  $\leq 0.076$ ), but a low-dimensional multi-ability structure generates much of it (rms 0.162); the observed spectral signature resembles the latter, not duplicated model families. The honest reading is therefore that open-model failure is governed by a handful of latent dimensions beyond unidimensional ability. We also reconcile two apparently contradictory statistics exactly — the excess variance is  $29\times$  the null while its covariance with the margin-model prediction is strongly negative — and show the effect is not an artifact of duplicate models, chain mixing, or the choice of spectral functional.

## 1 Introduction

If many deployed models share blind spots, diversification is illusory: an ensemble, a panel of model judges, or a market of competing screening vendors supplies far less independent evidence

than its headcount suggests. A growing literature measures how often models err together, and almost all of it shares one pattern: record which models fail which items, then compare the rate at which pairs fail together against a baseline in which they are assumed independent.

Jo et al. [2026] showed that this comparison is highly sensitive to the assumed baseline — conditioning on item difficulty leaves residual correlations “substantially attenuated”, with “some even flipping from strongly positive to slightly negative” — and proved that a sufficiently expressive null can explain away any observed dependence. Their conclusion is that monoculture is a context-dependent inference problem, and they state explicitly that their framework “does not prescribe which null is right”.

We take up that open question. Our claim is that the ladder of nulls has a canonical rung. The row and column margins of a binary outcome matrix are the jointly sufficient statistics of the Rasch model; conditioning on them therefore removes the entire nuisance family without estimating any of it. The resulting null — the uniform distribution over margin-matched matrices, known in ecology as the fixed-fixed null and in psychometrics as the Rasch conditional null — requires no gradient ascent, no regularisation, and no identification convention. It is the point at which “we conditioned on ability and difficulty” stops being an analyst’s choice and becomes an exact statement.

## Contributions.

1. **A canonical stopping point.** We identify the exact conditional null implied by Rasch sufficiency as the non-subjective answer to the null-selection problem posed by Jo et al. [2026], and report what the answer is there (Section 4).
2. **A translation table across three literatures.** The degeneracy of mean co-failure under margin-preserving nulls is Schluter’s V-ratio result in ecology [Schluter, 1984, Gotelli, 2000, Gotelli & Ulrich, 2012], Kuder–Richardson/Cronbach’s  $\alpha$  in psychometrics [Kuder & Richardson, 1937, Cronbach, 1951], and Fleiss’s observed agreement in the multi-category case [Fleiss, 1971]. None is cited in the model-evaluation literature. We state the identities and attribute them (Section 3).
3. **A calibrated measurement of residual structure at scale,** over 1,228–1,373 models per benchmark, together with a negative result about the statistic most tempting to report: the participation ratio is a monotone transform of mean squared residual correlation and does not count independent models (Section 6).
4. **An exact reconciliation of two contradictory-looking results,** and robustness to deduplication, mixing, and functional choice (Sections 6.6–6.7).
5. **A released substrate and estimator,** harvested at zero inference cost, with the harness pitfalls documented (Section 5).

**What we do not claim.** We do not claim prior measurements were computed incorrectly, nor that language models are in fact diverse. We claim that a class of baseline cannot separate shared behaviour from shared item difficulty; that a canonical baseline exists; and that at that baseline the concentration conclusion survives in a different and more defensible form than the one usually reported.

## 2 Setup

Let  $F \in \{0, 1\}^{N \times M}$ ,  $F_{im} = 1$  iff model  $i$  fails item  $m$ . Write  $r_i = \sum_m F_{im}$ ,  $c_m = \sum_i F_{im}$ ,  $p_i = r_i/M$ ,  $f_m = c_m/N$ ,  $\bar{f} = \frac{1}{M} \sum_m f_m$ , and  $C_{ij} = \frac{1}{M} \sum_m F_{im} F_{jm}$ .

### 3 The mean is degenerate, and this is known

**Lemma 1** (Mean co-failure depends only on item margins).  $O = \frac{1}{MN(N-1)} \sum_m (c_m^2 - c_m)$ .

*Proof.*  $\sum_{i \neq j} F_{im} F_{jm} = (\sum_i F_{im})^2 - \sum_i F_{im}^2 = c_m^2 - c_m$  since  $F_{im} \in \{0, 1\}$ .  $\square$

Consequently any randomisation preserving the column margins leaves  $O$  invariant with zero variance, and the standardised effect size for  $O$  is 0/0.

**This is not new, in three separate fields.** In ecology it is the degeneracy of Schluter’s V-ratio [Schluter, 1984]. Gotelli [2000] states plainly that “the V ratio was not used in this analysis because retaining row and column totals as in SIM9 does not ever change the index”, and Gotelli & Ulrich [2012] generalise the warning to the Sørensen, Simpson, Morisita and NODF families. In psychometrics the same content appears as reliability:

**Lemma 2** (The independence-baseline excess is reliability times difficulty variance). *With  $I$  the mean of  $p_i p_j$  over ordered pairs,*

$$O - I = \frac{N \text{Var}_m(f_m) + \text{Var}_i(p_i) - \bar{f}(1 - \bar{f})}{N - 1} = \alpha_N \cdot \text{Var}_m(f_m),$$

where  $\alpha_N$  is Cronbach’s  $\alpha$  computed with the  $N$  models in the item role [Kuder & Richardson, 1937, Cronbach, 1951].

We verified the second equality numerically: residual  $\leq 6.9 \times 10^{-17}$  across five shapes with  $N \in [25, 1200]$ ,  $M \in [45, 900]$ , and  $-4.2 \times 10^{-17}$  on the real ARC matrix, where  $\alpha_N = 0.99928$ . Since  $\alpha_N \rightarrow 1$  for large  $N$ , the reported “excess co-failure” is, to an excellent approximation, simply the cross-item variance in difficulty — strictly positive whenever items differ in difficulty, including when every model fails independently. The asymptotic statement is the de Finetti / beta-binomial identity, and Yule [1903] is the classical warning about association induced by mixing.

For multi-category responses the analogous statistic — the rate at which two models give the same wrong answer given both are wrong, as used by Kim et al. [2025] against a uniform  $1/(K-1)$  baseline — is Fleiss’s observed agreement [Fleiss, 1971],  $\sum_m \sum_k n_{mk}(n_{mk} - 1) / \sum_m n_m(n_m - 1)$ , and depends on the per-item response composition alone. In a simulation where models err *conditionally independently* given the item, drawing from a shared per-item distractor-attractiveness distribution, we observe agreement 0.5567 against a uniform baseline of 0.3333 — an apparent excess of +0.2233 — while the excess over the composition-preserving null is  $-1.1 \times 10^{-16}$ . This is the founding premise of answer-copying detection [Wollack, 1997, van der Linden & Sotaridona, 2006], and we restate it rather than claim it.

**Why restate known results.** Because the model-evaluation literature cites none of them, and because the three statements are the same theorem in three notations. Making the bridge explicit is the cheapest available correction to a measurement practice now in active use.

### 4 The canonical null

Jo et al. [2026] prove that any distribution on  $\{0, 1\}^m$  is a mixture of conditionally independent Bernoullis, so excess dependence shrinks monotonically as the null grows more expressive; they conclude the choice is subjective. We observe that the ladder has a distinguished rung.

The Rasch model  $\Pr(F_{im} = 1) = \sigma(\alpha_i + \beta_m)$  has  $\{r_i\}$  and  $\{c_m\}$  as jointly sufficient statistics. Conditioning on both therefore eliminates  $\alpha$  and  $\beta$  exactly, and the conditional law of  $F$  given

| Benchmark        | Models | Items | Mean accuracy | Snapshots         |
|------------------|--------|-------|---------------|-------------------|
| ARC-Challenge    | 1362   | 1165  | 0.527         | 2023-07 – 2024-06 |
| Winogrande       | 1361   | 1267  | 0.734         | 2023-09 – 2024-06 |
| TruthfulQA (mc1) | 1334   | 786   | 0.368         | 2023-07 – 2024-06 |
| GSM8K            | 1228   | 1319  | 0.348         | 2023-09 – 2024-06 |

Table 1: Substrate after removing degenerate rows and columns. For comparison, Jo et al. [2026] analyse 451 models, Kim et al. [2025] 349.

its margins is *uniform* on the set of margin-matched binary matrices, for every parameter value [Rasch, 1960, Andersen, 1973]. That set is precisely the ecologists’ fixed-fixed null, sampled here with curveball [Strona et al., 2014]; the equivalence between the two traditions is standard [Besag & Clifford, 1989, Ponocny, 2001, Verhelst, 2008, Miller & Harrison, 2013].

The practical consequence is that this null estimates nothing. A fitted IRT null requires a parameterisation, an optimiser, penalties, and an identification convention, each of which is a defensible-but-arbitrary analyst choice of exactly the kind Jo et al. [2026] identify. The conditional null has none. It is the strongest ability-and-difficulty control that can be applied without fitting, and it is unique in that respect.

**Margin-conditioned excess.** For a point estimate of the null mean we use the maximum-entropy (Rasch) mean field  $P_{im} = \sigma(\alpha_i + \beta_m)$ , fitted to a maximum absolute margin error of  $5.7 \times 10^{-13}$ , and

$$D = \frac{FF^\top - PP^\top}{M}.$$

The mean field is validated against the randomisation itself: over 319,600 model pairs the curveball expectation of  $C_{ij}$  and the Rasch prediction correlate at 0.99999, mean absolute difference 0.00039 on a level of 0.2849. The matrix  $D$  is the excess matrix of the network-backbone literature [Neal et al., 2021].

**Definition 1** (Effective number of independent models). Let  $R$  be  $D$  scaled to unit diagonal, with eigenvalues  $\lambda_k$ . Then  $N_{\text{eff}} = (\sum_k \lambda_k^+)^2 / \sum_k (\lambda_k^+)^2$ .

The participation ratio, and the practice of removing a dominant shared mode before reading the bulk, are standard in econophysics [Laloux et al., 1999, Plerou et al., 1999]; the same quantity is  $M_{\text{eff}}$  in statistical genetics [Cheverud, 2001, Nyholt, 2004, Li & Ji, 2005]; eigen-decomposing Rasch residuals is standard psychometrics [Linacre, 1998]. Our contribution here is the calibration —  $N_{\text{eff}}$  must be compared against exact-margin replicates, because its null value is emphatically not  $N$  — and the application to model redundancy.

## 5 Data

We harvest per-item outcomes from the archived Open LLM Leaderboard. Only the metric column of each file is read — 1.4 KB out of a 63 MB file for HellaSwag — which makes an ecosystem-scale harvest possible at zero inference cost.

Two pitfalls had to be handled and are worth recording. Item identifiers are not stable: the harness stored a dataset identifier in mid-2023 and the raw question text thereafter, so a naive identifier join returns an *empty* intersection. The accuracy field also moved into a nested **metrics** struct in the 2024 schema. We resolve item identity by row index, licensed by explicit verification — row order was identical for 14 of 14 sampled models spanning July 2023 to May 2024 and all three schema generations — and enforce a row-count guard on every read.

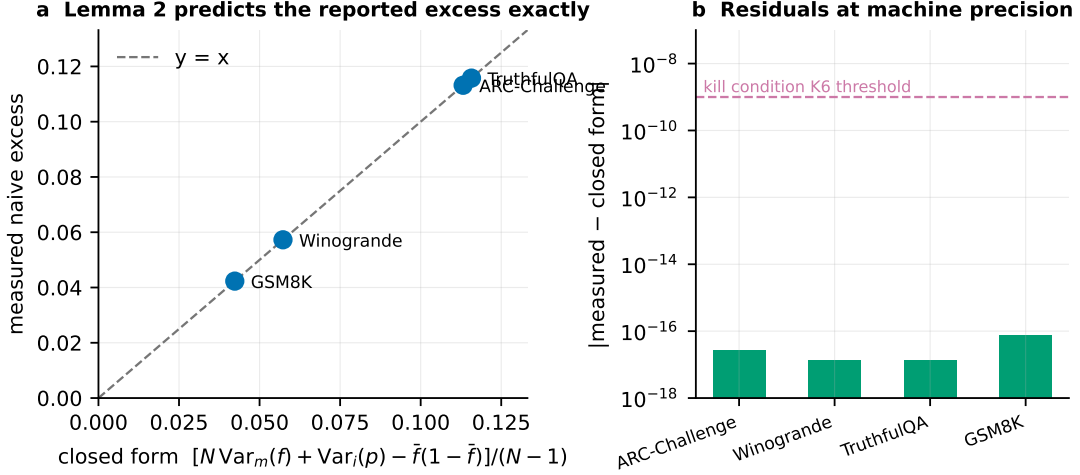


Figure 1: Lemma 2 predicts the measured excess exactly on all four benchmarks (panel a); the residual sits at machine precision, well below the  $10^{-9}$  validation threshold set in the pre-registration (panel b).

| Benchmark     | $T$ observed           | $T$ null               | SES    | ratio |
|---------------|------------------------|------------------------|--------|-------|
| ARC-Challenge | $7.640 \times 10^{-3}$ | $8.331 \times 10^{-3}$ | -70.9  | 0.917 |
| Winogrande    | $1.310 \times 10^{-3}$ | $2.322 \times 10^{-3}$ | -149.8 | 0.564 |
| TruthfulQA    | $1.277 \times 10^{-2}$ | $1.164 \times 10^{-2}$ | +80.1  | 1.098 |
| GSM8K         | $4.576 \times 10^{-2}$ | $4.623 \times 10^{-2}$ | -38.5  | 0.990 |

Table 2: Independent-chain nulls,  $R = 200$  chains each burned in 50 trades per model.

## 6 Results

### 6.1 The closed form is exact on real data

Measured excess versus Lemma 2: ARC  $+0.113180719$  vs  $+0.113180719$  (residual  $-2.8 \times 10^{-17}$ ); Winogrande  $+0.057265401$  ( $-1.4 \times 10^{-17}$ ); TruthfulQA  $+0.115767366$  ( $+1.4 \times 10^{-17}$ ); GSM8K  $+0.042311309$  ( $+7.6 \times 10^{-17}$ ). Over 400 curveball replicates per benchmark, with both margins exactly preserved, the maximum deviation of mean co-failure from its observed value is  $0.00 \times 10^0$ , as Lemma 1 requires. Of a reported excess as large as  $+0.116$ , none survives conditioning.

### 6.2 The pre-registered dispersion statistic is refuted

Our pre-registered primary test was whether  $T = \text{Var}_{i < j}(C_{ij})$  exceeds its null. It does not do so consistently (Table 2); the pre-registered kill condition required the effect in at least two of three primary benchmarks and it fired. We report  $T$  as uninformative rather than reinterpreting it.

### 6.3 What survives: effective independence

### 6.4 Why these numbers are not an “effective number of models”

It is tempting to read Table 3 as “1,362 models behave like 24”. That reading is wrong, and the reason is algebraic. For any unit-diagonal  $R$ ,

$$\text{PR} = \frac{(\sum_k \lambda_k)^2}{\sum_k \lambda_k^2} = \frac{N}{1 + (N-1) \overline{R_{ij}^2}},$$

| Benchmark     | $N$  | $N_{\text{eff obs.}}$ | $N_{\text{eff null}}$ | ratio | raw (unconditioned) |
|---------------|------|-----------------------|-----------------------|-------|---------------------|
| ARC-Challenge | 1362 | 23.6                  | $449.3 \pm 3.4$       | 0.052 | 1.9                 |
| Winogrande    | 1361 | 20.1                  | $493.8 \pm 3.0$       | 0.041 | 3.6                 |
| TruthfulQA    | 1334 | 17.4                  | $309.7 \pm 2.4$       | 0.056 | 1.5                 |
| GSM8K         | 1228 | 118.6                 | $553.1 \pm 2.2$       | 0.214 | 2.1                 |

Table 3: The last column shows why calibration is not optional: uncalibrated, the participation ratio reads 1.5–3.6, which is not a finding but the single shared difficulty eigenvalue.

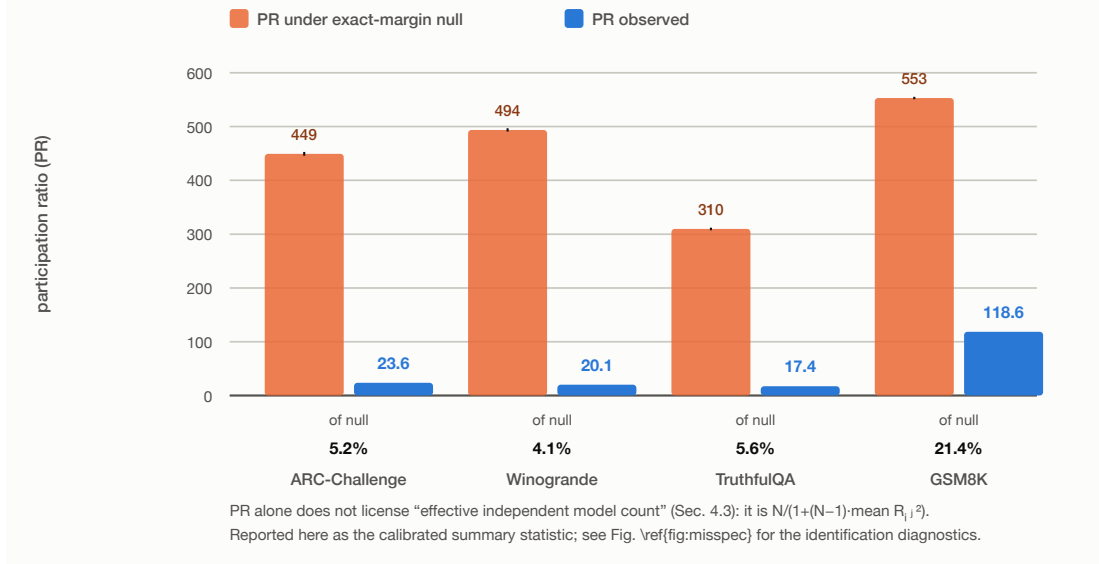


Figure 2: Observed participation ratio against its exact-margin null, all four benchmarks. Read this alongside Section 6.4: PR is reported here as a calibrated summary statistic, not as a count of independent models.

which we confirm to machine precision on ARC (16.0449 computed either way). The eigendecomposition contributes nothing: the participation ratio is a monotone transform of the mean squared off-diagonal residual correlation. An equicorrelated matrix — one weak global factor plus  $N - 1$  independent directions — reproduces the reported value exactly, and so does a partition into 24 blocks of exact clones. The statistic cannot tell them apart, so it cannot license a count of independent models.

The diagnostics that can tell them apart give a clear answer on ARC:  $\lambda_1^2 / \sum \lambda_k^2 = 0.540$  (one-factor  $\approx 0.99$ , clone blocks  $\approx 0.10$ ); deflating the leading eigenvector raises PR only from 16.0 to 46.8, not to  $\approx N$ ; and five observed eigenvalues exceed the null’s maximum eigenvalue of 28.7. We therefore report the primitive quantity,  $\text{rms}|R_{ij}| = 0.2483$  against 0.0389 under the null, and the count of dimensions above the null spectral edge.

## 6.5 Misspecification controls: what else could produce this

Our conditioning model is one-parameter. Any two-parameter or multidimensional structure leaves residual signal that the margin-preserving null cannot reproduce, so we ask how much of the effect that alone explains. Each row below is a simulated population with *no* clusters, no clones and no shared lineage — errors conditionally independent given the item.

Two conclusions follow. Discrimination heterogeneity is *not* a sufficient explanation: even at  $\text{sd}(\log a) = 0.60$  it yields rms 0.076 against the observed 0.248. But a small number of extra ability dimensions is a substantial one, and the observed  $\lambda_1^2 / \sum \lambda^2 = 0.540$  sits near the two-dimensional case (0.457) and far from the clone case (0.102). We therefore state the

| Generating process       | PR / PR <sub>null</sub> | rms $ R_{ij} $ | $\lambda_1^2 / \sum \lambda^2$ | dims > edge |
|--------------------------|-------------------------|----------------|--------------------------------|-------------|
| 1PL, correctly specified | 1.001                   | 0.065          | 0.105                          | 1           |
| 2PL, sd(log $a$ ) = 0.35 | 0.966                   | 0.068          | 0.119                          | 1           |
| 2PL, sd(log $a$ ) = 0.60 | 0.862                   | 0.076          | 0.167                          | 1           |
| two ability dimensions   | 0.230                   | 0.162          | 0.457                          | 3           |
| 20 exact clone clusters  | 0.112                   | 0.234          | 0.102                          | 19          |
| <b>real ARC</b>          | <b>0.052</b>            | <b>0.248</b>   | <b>0.540</b>                   | <b>5</b>    |

Table 4: Discrimination heterogeneity alone cannot produce the observed effect. A low-dimensional multi-ability structure produces much of it, and the observed spectral signature resembles that case rather than duplicated model families.

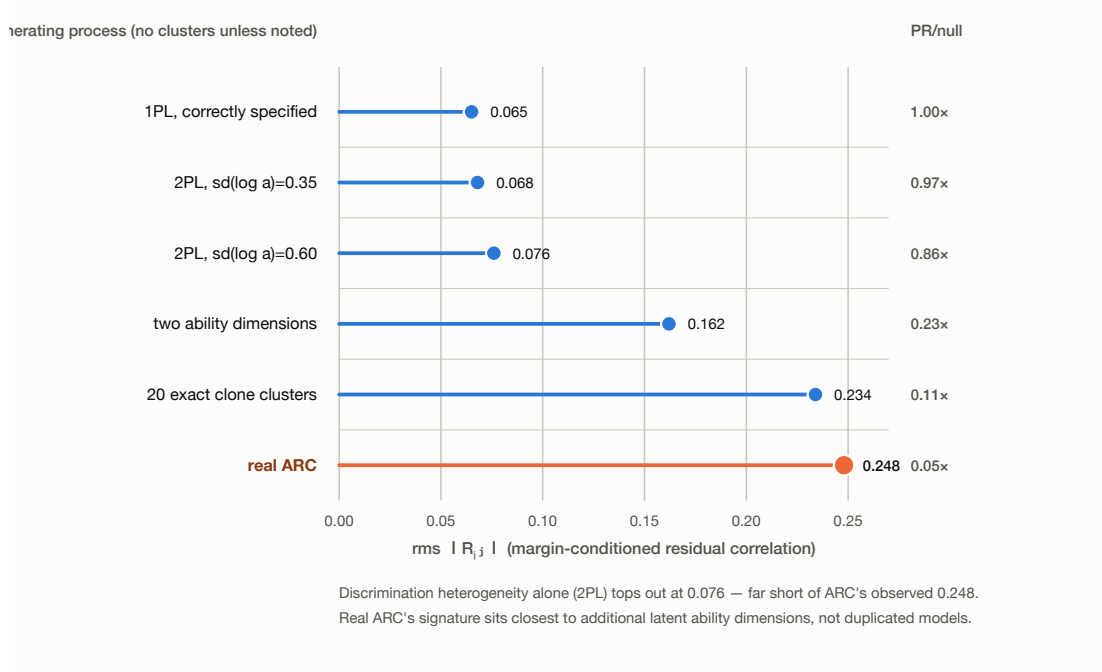


Figure 3: Identification evidence: real ARC's residual correlation (orange) sits far beyond what two-parameter discrimination heterogeneity alone can produce, and its  $\lambda_1^2 / \sum \lambda^2$  signature (Table 4) matches a multi-ability structure rather than duplicated models.

finding as: after conditioning on item difficulty and unidimensional ability, open-model failure retains correlated structure of roughly five dimensions and sixfold excess rms correlation, whose signature is that of additional latent ability dimensions rather than of duplicated models.

## 6.6 Reconciling two opposite-looking results

$T$  below null and  $N_{\text{eff}}$  far below null appear to conflict: since  $R$  has unit diagonal,  $\text{tr } R = N$  exactly and  $N_{\text{eff}} = N^2 / \|R\|_F^2$ , so  $N_{\text{eff}}$  far below null is algebraically equivalent to the off-diagonal excess correlations being far *above* null. The resolution is exact. Because the null preserves margins, the fitted  $P$  — and hence  $E = PP^T / M$  — is identical for the observation and every replicate; we measure  $|\text{Var}(E)_{\text{obs}} - \text{Var}(E)_{\text{null}}| = 0.000 \times 10^0$ . Writing  $C = E + D$ ,

$$\text{Var}(C) = \text{Var}(E) + 2 \text{Cov}(E, D) + \text{Var}(D).$$

On ARC:  $\text{Var}(D) = 6.253 \times 10^{-4}$  against a null of  $2.127 \times 10^{-5}$  (SES +1613, a 29x excess), while  $\text{Cov}(E, D) = -6.494 \times 10^{-4}$  against  $-2.1 \times 10^{-6}$  (SES -153; correlation -0.285 versus

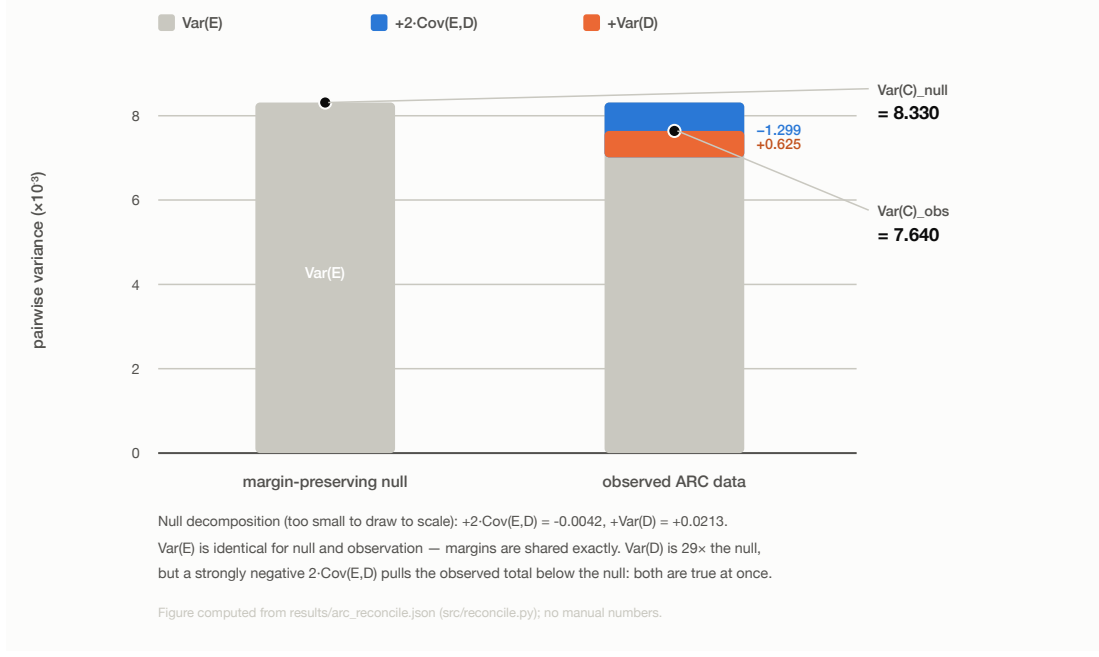


Figure 4: The exact decomposition  $\text{Var}(C) = \text{Var}(E) + 2\text{Cov}(E, D) + \text{Var}(D)$  on ARC.  $\text{Var}(E)$  is identical for null and observation by construction; the observed column's negative covariance term outweighs its 29 $\times$ -inflated variance term, landing below the null.

−0.005). The negative covariance outweighs the excess variance, which is exactly why  $T$  reads below null. Both statistics are correct and the arithmetic closes to the reported  $\text{Var}(C)$ .

Substantively: pairs that the margin model predicts will co-fail most actually co-fail relatively less, and vice versa. That is structured deviation from unidimensionality — which is what “models cluster into a few dozen behavioural types” means when stated in Rasch terms. We flag that “residual structure” and “Rasch misfit” are the same observation under two descriptions, and that our design cannot separate them.

## 6.7 Robustness

**Near-duplicates.** The archive contains exact duplicates (maximum pairwise agreement 1.0000). Clustering models by agreement and keeping one representative per cluster: on ARC, removing 900 of 1,362 models moves  $N_{\text{eff}}$  from 23.6 only to 29.0; on TruthfulQA, removing 1,001 of 1,334 moves it from 17.4 to 26.2; on GSM8K, removing 450 moves it from 118.6 to 111.4. Redundant submissions are not the mechanism.

**Mixing.**  $T$  plateaus by about 10 trades per model and is flat to 200. Independent chains give null SD ratios 0.96–1.02 versus single-chain thinning, with lag-1 autocorrelations between −0.14 and +0.07. A 2000-trades-per-model burn-in gives an ARC  $N_{\text{eff}}$  null of 450.8 against 449.3 at 50 — a 40 $\times$  longer burn-in moves it by 0.3%. Both margins are exactly preserved at every checkpoint.

**Functional choice.** On ARC, 91.35% of spectral mass is positive. The participation ratio is insensitive to the treatment of negative eigenvalues: 23.6 clipped, 23.5 absolute, 16.0 raw.

**Estimator controls.** On a matrix drawn from a Rasch margin model the dispersion estimator reports SES +0.05; on eight latent model families +6.75; on exact clones +2.64; on column-shuffled real data −1.33.



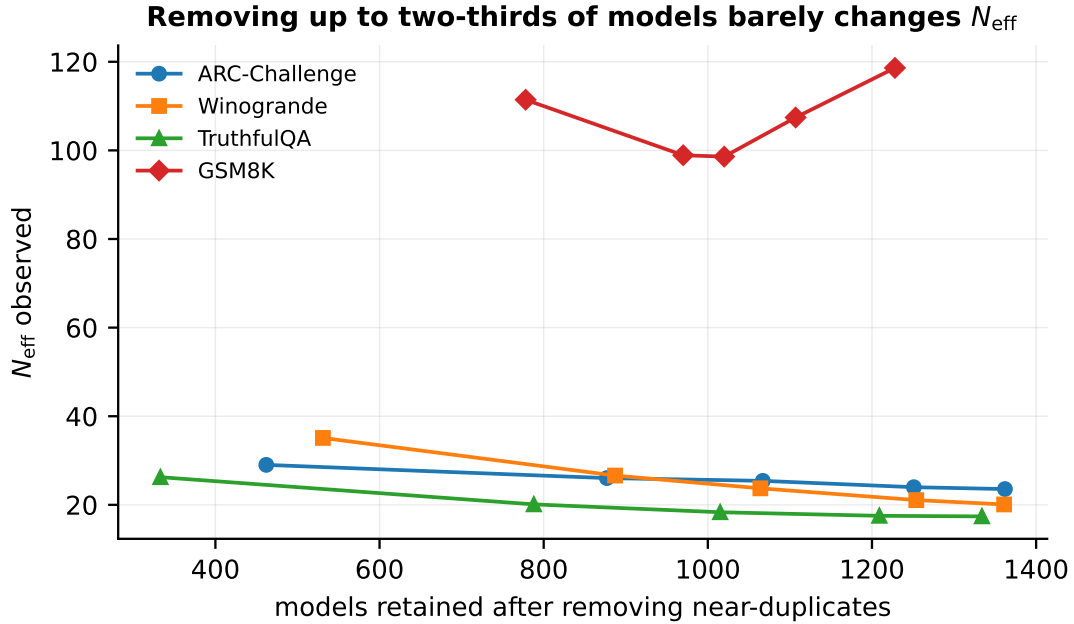


Figure 5: Observed participation ratio as near-duplicate models (agreement  $\geq$  threshold, one representative kept per cluster) are progressively removed. Removing up to two-thirds of each benchmark’s models moves the statistic by only a few units.

## 7 Related work

Kim et al. [2025] measure agreement-when-both-wrong across 349 models against a uniform baseline, conditioning on model accuracy but not item difficulty, and report that larger and more accurate models have more correlated errors. Jo et al. [2026] show that conditioning on item difficulty via fitted IRT attenuates such correlations substantially and sometimes reverses their sign, and argue the inference is null-dependent and hence subjective; our contribution is to identify a null on that ladder which requires no fitting, and to report the answer there. Foundational framing of algorithmic monoculture and outcome homogenisation is due to Kleinberg & Raghavan [2021] and Bommasani et al. [2022]. On method, the fixed-fixed null and curveball come from ecology [Connor & Simberloff, 1979, Schluter, 1984, Gotelli, 2000, Strona et al., 2014]; the equivalent Rasch conditional machinery from psychometrics [Rasch, 1960, Andersen, 1973, Besag & Clifford, 1989, Ponocny, 2001, Verhelst, 2008, Miller & Harrison, 2013]; the participation-ratio and shared-mode-removal apparatus from econophysics [Laloux et al., 1999, Plerou et al., 1999] and statistical genetics [Cheverud, 2001, Nyholt, 2004, Li & Ji, 2005]; and the excess-matrix construction from network backbone extraction [Neal et al., 2021].

## 8 Limitations

Claims cover the evaluated open-model ecosystem on multiple-choice-style benchmarks, not deployed proprietary systems; the societal framing is motivation, not a measured outcome, and we make no claim about real-world screening decisions. The models are a self-selected population of leaderboard submissions, and Jo et al. [2026] show explicitly that such inferences are population-dependent — a different peer set can change them.

The central limitation is identification. Our conditioning model is unidimensional, and Table 4 shows that additional latent ability dimensions reproduce much of the observed effect with no shared lineage whatsoever. We can rule out discrimination heterogeneity as a sufficient

explanation, and we can rule out duplicate models, but we cannot separate “genuine behavioural concentration” from “the Rasch family is too small” — and on the spectral evidence the latter is the better description. Re-conditioning on a fitted 2PL or low-rank logistic model would sharpen this at the cost of the exactness that motivates the design; that trade-off is the natural next step.

We also report a negative methodological result about our own pre-registered plan: the participation ratio does not measure an effective number of independent models (Section 6.4), and we have removed that interpretation rather than retain it.

Pre-registered kill condition K5 failed — declared `base_model` coverage was 23.3% against a 40% threshold, with 315 of 1,362 model repositories no longer retrievable — so the lineage decomposition is dropped rather than estimated on a biased subset. The multi-category result is established analytically and in simulation; a real-response analysis must be explicit about the estimator, since the pooled agreement ratio is composition-invariant but the pair-averaged form is not.

## 9 Reproducibility and use of AI assistance

All code, the harmonised substrate, and every results artifact are released; each number traces to a JSON file emitted by an executed run, and the propositions are asserted as unit tests. The study was pre-registered before the confirmatory analysis, with two dated amendments recorded before any confirmatory statistic was computed; one pre-registered hypothesis was refuted by its own kill condition and is reported as refuted.

This work used a large language model (Claude, Anthropic) as a coding and analysis assistant: it wrote the harvesting, estimation and figure code and assisted in drafting and in the adversarial prior-art review that led to Section 3 being rewritten from claimed novelty to attribution. The author directed the research, specified and pre-registered the hypotheses and kill conditions, verified the derivations, and takes full responsibility for the content, in line with COPE guidance that AI tools cannot be authors.

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